AHMED MAHFOOZ ALI KHAN

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TECHNICAL SKILLS

Languages: C, C++, C#, SQL, Java, HTML, CSS, .NET, Matlab, LaTeX, Rust, English, Swedish
Libraries: Pandas, NumPy, React, OpenCV, Matplotlib
ML/NLP Frameworks: TensorFlow, Keras, Scikit-learn, RAG, NLTK, spaCy, LightGBM, Ollama, LangChain,
Statistical Analysis/Deep Learning: Regression Analysis, Hypothesis Testing, Clustering, RNNs, GNNs, CNNs
Tools: Maltego, Nmap, Sqlmap, Burp Suite, Hydra, Rightway, Rheino
Network Protocols: HTTP, FTP, SSH, SMTP, DNS, DHCP, TCP/IP
Vulnerability Assessment: Nessus, Nexpose, Nikto, Wireshark, Splunk
Operational Security:Vulnerability & patch management support, Linux hardening, compliance checks, baselines handling, runbooks, ticket traceability, remediation validation, dashboards/KPIs, Bash automation
Methodologies: OWASP Top 10, SANS Top 25, SDLC, MITRE ATT&CK
Operating Systems: Windows, Mac OS, Linux, Parrot, Ubuntu, Git

EDUCATION

Blekinge Institute of Technology — Karlskrona, Sweden
Master's Degree in Computer Science — January 2024 – February 2026

Jawaharlal Nehru Technological University — Hyderabad, India
Bachelor's Degree in Computer Science Engineering — December 2020 – December 2023

EXPERIENCE

Artificial Intelligence Master Thesis — December 2024 – June 2025
Volvo Construction Equipment — Gothenburg, Sweden

- Spearheaded a cutting edge research initiative under the SUGAR program in partnership with (BTH) and Volvo CE, focusing on leveraging AI methodologies to optimize construction site workflows and performance.
- Devised an AI-driven service and business model that cut downtime by 20% and improved on site asset integration; built a RAG prototype with PostgreSQL for retrieval backed decision support.
- Presented project advancements during an international summit in São Paulo, Brazil, engaging stakeholders and incorporating feedback from leading industry experts.
- Showcased final deliverables at Silicon Valley, USA, demonstrating the practical implications and scalability of AI-enabled solutions to a global audience of researchers and practitioners.

Artificial Intelligence Internship — July 2024 – September 2024
Price Waterhouse Coopers International Limited (PwC) — Zurich, Switzerland

- Completed a job simulation involving 25% data analysis and modelling for the Artificial Intelligence team.
- Built Python classification models, improving data driven decision accuracy by 30% using GenAI technology.
- Analyzed 60% data for adverse effects, compared medications and contributed to a client data strategy proposal.
- Refined data processes through advanced feature engineering and robust statistical analyses, ensuring the precision and applicability of predictive model outputs.

Cyber Security Internship — May 2024 – July 2024
Entersoft Labs — Hyderabad, India

- Conducted Behavioral Analysis, System Testing on web applications, reducing potential attack vectors by 25%.
- Monitored network traffic using SIEM tools, Log Analysis contributing to early threat detection by 40%.
- Configured firewalls, IDS/IPS and antivirus systems, enhancing network security measures by 30%.
- Utilized a range of advanced security tools, including Burp Suite, OWASP ZAP and Metasploit, to assess vulnerabilities and strengthen application defenses by 25%.

Full Stack Web Development Internship — October 2023 – December 2023
Indian Institute of Technology (IIT) — Kanpur, India

- Engineered a robust API using Node.js, Express.js and MongoDB, improving reliability and reducing response time by 40% through rigorous pre deployment testing; containerized with Docker and deployed to a Kubernetes environment.
- Implemented user authentication and authorization using JWT, reducing unauthorized access incidents by 50%.
- Conceptualized and implemented an advanced Front End Web UI leveraging CSS, JavaScript, and React, resulting in a 25% enhancement in user engagement through optimized design and functionality.
- Deployed containerized services to Azure via CI/CD and Terraform for reproducible environments.

PROJECTS

Character Encoding and Image Decoding | *PyTesseract, PIL, PyTorch* September 2024 – November 2024

- Conceived an image encoding function boosting barcode conversion accuracy by 23%, ensuring precise representation.
- Leveraged Python image processing libraries to optimize execution speed by 30%, enhancing performance & reliability.
- Crafted a decoding algorithm increasing pixel data parsing accuracy by 29%, strengthening text recognition accuracy.

Gravity Anomaly Prediction Using Machine Learning | *Scipy, Xgboost* March 2025 – May 2025

- Tested 13 machine learning models, including KNN, Random Forest, neural networks, and graph-based techniques.
- Established that nonlinear models surpassed linear ones by over 20% in performance, highlighting the suitability of nonlinear methods for this application.
- Recorded an R^2 of 99.7% and a MAE of 1.09 mGal with the Hybrid KNN+RF model, demonstrating precise predictions for geophysical datasets.

AR Gesture Controlled Threat Shooter | *MediaPipe, OpenCV* January 2026 – March 2026

- Orchestrated 100% gesture based gameplay with real time aiming and flick based shooting.
- Accelerated gameplay intensity by 60% by reducing spawn intervals and scaling threat difficulty over time.
- Stabilized aiming precision by up to 70% through adaptive smoothing for steadier and more responsive control.

Digitalization of Traffic Management | *JavaScript, Pandas, HTML, Matplotlib* March 2024 – May 2024

- Developed an optimized pre-gate parking system utilizing 3D Lidar surveillance and monitoring, enhancing parking layout efficiency by 30%, reducing congestion, and minimizing vehicle wait times.
- Created an adaptive decision support system (DSS) that increased port traffic management capacity by 25%, ensuring smooth traffic flow and compliance with safety regulations.
- Enhanced overall port operational efficiency by reducing parking congestion by 20%, utilizing machine learning and predictive analytics to streamline vehicle navigation and parking slot allocation.

Teacher - Student Knowledge Distillation for Model Compression | *Keras* August 2025 – September 2025

- Devised a teacher-student knowledge distillation strategy in TensorFlow, compressing model size by 12% without compromising predictive accuracy.
- Formulated optimal temperature scaling and distillation loss-weighting, driving a 15% reduction in inference latency for deployment ready models.
- Curated and preprocessed CIFAR-10 datasets, delivering a 9% accuracy boost in the distilled student network via enhanced augmentation pipelines.

SECURING IoT DATA | *HTTPS, STRIDE, NumPy, SHA256* June 2023 – December 2023

- Programmed a secure IoT data solution with blockchain integration in the cloud, enhancing data integrity and transparency by 75%, while incorporating anomaly detection for real-time threat identification.
- Implemented cryptographic permissions to enhance data privacy, achieving 90% of data ownership.
- Lowered system failure risks by 80% using blockchain decentralized architecture combined with predictive analytics.

CERTIFICATIONS

- EC-Council Certified Ethical Hacker (CEH)
- Microsoft Certified Natural Language Processing in Microsoft Azure
- JP Morgan Certified Software Engineer

ACHIEVEMENTS

- Honored with a Gold Medal in a running event during a cancer awareness campaign.
- Awarded 50% Merit Scholarship for master's degree in Computer Engineering at BTH, Sweden.
- Authored a research paper titled "A Steganographic Approach for Medical Imaging" Enhancing Image Quality and Security, demonstrating expertise in advanced imaging techniques and data security.
- Published an research paper "AI-driven Optimization Framework for Construction Site Ecosystems", that explores predictive analytics and real-time decision making to enhance equipment coordination and site wide efficiency.

EXTRACURRICULAR

- Recognized as the Best Team Management Leader at the Job Mela hosted by JNTUHCEH, exemplifying exceptional leadership, team coordination, and organizational expertise.
- Managed audience coordination for an Indian cultural event at Karlskrona Konserthusteater, ensuring smooth entry, seating, engagement and exit throughout the event.
- Representing my institution by conducting a Master's thesis presentation at **SAP, SFU and Stanford**, fostering cross institutional collaboration.